Bachelor of Computer Applications

Semester – V

Business Analytics

Experiment No.: 5

Title: Building Customer analysis Dashboard

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Management wants to quickly understand sales performance across products, categories, and regions. Create appropriate Power BI visualizations to identify key business patterns.

GitHub Link : https://github.com/athul-hue/BUSINESS-ANALYTICS-PROJECT/blob/main/Customer_Analysis%205.pbix

1. Problem Statement

Businesses need to understand customer purchasing behaviour, sales performance, and revenue trends to make better decisions. Analysing customer transactions helps identify popular product categories, preferred payment methods, high-value customers, and revenue concentration.

The objective of this project is to develop an interactive Customer Analysis Dashboard using Microsoft Power BI to analyse customer and order data and present meaningful business insights through visualizations.

2. Objectives

1. Identify the most popular product categories.
2. Analyse the most frequently used payment methods.
3. Track monthly revenue trends.
4. Identify customers who make the most purchases.
5. Determine whether a small number of customers contribute a large proportion of total revenue.
6. Present important business metrics through an interactive dashboard.

3. Dataset Description

The project uses two CSV datasets:

- **Customer Dataset:** Contains customer information, including Customer ID, Customer Name, City, State, and Customer Segment.

- **Orders Dataset:** Contains order information, including Order ID, Customer ID, Order Date, Product Category, Payment Method, Quantity, Unit Price, Discount Percentage, and Sales Amount.

The datasets were imported into Power BI Desktop for data preparation, modelling, and visualization.

4. Data Import and Preparation

The following steps were performed:

1. Imported both CSV files into Power BI Desktop.
2. Checked column names and data types.
3. Converted the order date into the appropriate date format.
4. Verified the numerical fields used for revenue and purchase analysis.
5. Created a relationship between the customer and orders tables using Customer ID.
6. Created a Calendar table to support monthly revenue analysis.
7. Developed DAX measures for calculating total revenue, total orders, purchasing customers, average order value, and customer purchase counts.

These steps prepared the data for analysis and visualization.

5. DAX Measures

The following DAX measures were created for the dashboard.

Total Revenue

```
Total Revenue =
SUM('customer_analysis_orders_5000'[SalesAmount])
```

Total Orders

```
Total Orders =
DISTINCTCOUNT('customer_analysis_orders_5000'[OrderID])
```

Purchasing Customers

Purchasing Customers =

DISTINCTCOUNT('customer_analysis_orders_5000'[CustomerID])

Average Order Value

Average Order Value =

DIVIDE(

    [Total Revenue],

    [Total Orders],

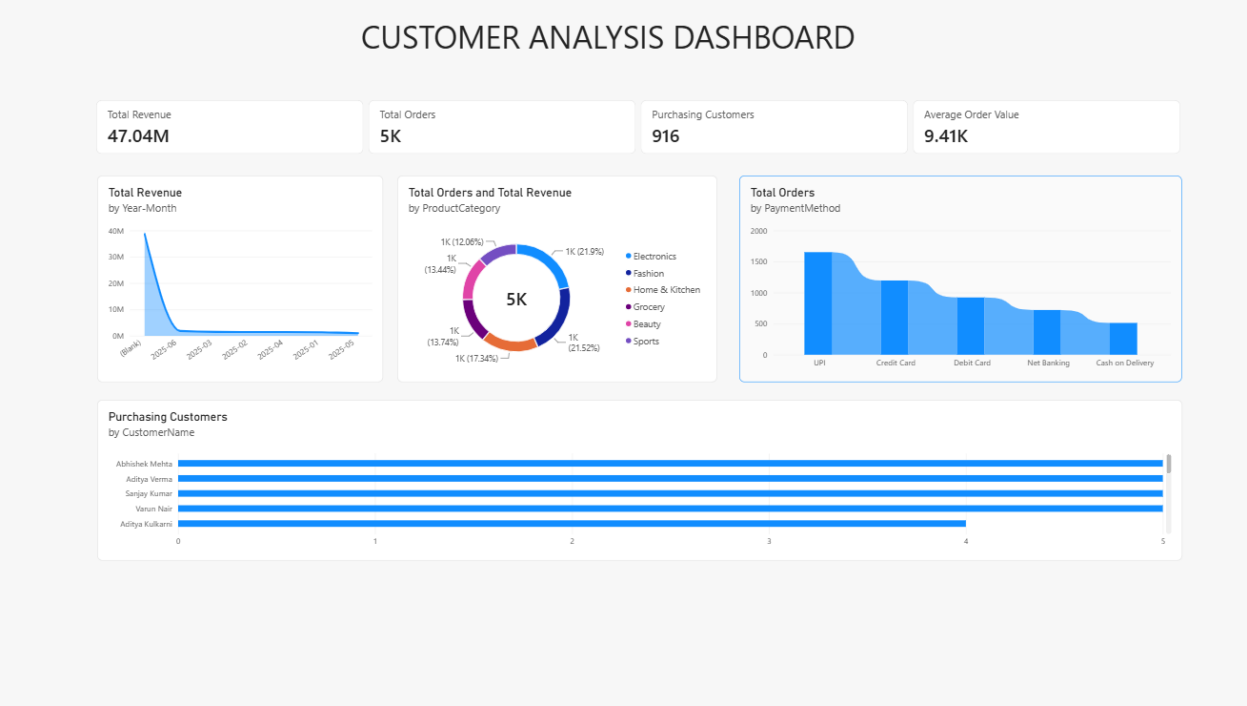
    0

)

Customer Purchase Count

Customer Purchase Count =

DISTINCTCOUNT('customer_analysis_orders_5000'[OrderID])



These measures provide the main metrics required to understand customer purchasing behaviour and sales performance.

6. Dashboard Development

The Customer Analysis Dashboard was developed in Power BI using Card visuals, a line chart, bar charts, and interactive report features.

6.1 KPI Cards

Four Card visuals were created to display the key performance indicators.

KPI	Dashboard value
Total Revenue	47.04M
Total Orders	5K
Purchasing Customers	916
Average Order Value	9.41K

These cards provide a quick overview of overall revenue, order volume, purchasing customers, and average revenue per order.

6.2 Monthly Revenue Trend

A line chart was created to display total revenue by Year-Month.

Purpose: To understand changes in revenue over time and identify periods of higher or lower sales.

The Calendar table was used to support date-based analysis. The month labels should be sorted chronologically, and any blank date category should be checked to ensure the visual displays the correct monthly sequence.

6.3 Product Category Analysis

A bar chart was created using Product Category and order-related measures.

Purpose: To compare product categories and identify which categories generate the greatest purchasing activity.

The dashboard displays Electronics, Fashion, Home & Kitchen, Grocery, Beauty, and Sports.

Electronics appears to have the highest value among the displayed categories, indicating strong purchasing activity in this category.

6.4 Payment Method Analysis

A column chart was created to compare the number of orders across payment methods.

The chart includes UPI, Credit Card, Debit Card, Net Banking, and Cash on Delivery.

The dashboard indicates that UPI is the most frequently used payment method, followed by Credit Card and Debit Card.

This analysis helps the business understand customer payment preferences.

6.5 Customer Purchase Analysis

A horizontal bar chart was created to display purchasing customers and their purchase counts.

Purpose: To identify customers who place the greatest number of orders.

The displayed customer names include Varun Nair, Aditya Kulkarni, Aditya Mehta, Ashwarya Iyer, and Harish Nair.

The chart shows that the highest displayed purchase count is five orders, while several other displayed customers have four orders each.

This information helps identify frequent purchasers who may be suitable for customer retention and loyalty initiatives.

6.6 Customer Revenue Concentration

Customer revenue concentration can be analysed using total revenue by customer and the percentage of revenue generated by the top 10 customers.

This analysis helps determine whether the business depends heavily on a small group of customers.

The Top 10 Customer Revenue Share measure can be displayed in a Card visual, while a bar chart can compare revenue generated by the highest-revenue customers.

The measure should be used to assess the contribution of these customers relative to total revenue.

7. Key Business Insights

Based on the dashboard:

1. Overall Revenue: The business has generated approximately 47.04 million in recorded revenue.
2. Order Volume: Approximately 5,000 orders are represented in the dashboard.
3. Purchasing Customers: There are 916 distinct customers with recorded purchases.
4. Average Order Value: The average revenue per order is approximately 9.41K.
5. Product Category: Electronics appears to have the highest value among the displayed categories.
6. Payment Preference: UPI is the most frequently used payment method.

7. Customer Purchasing Behaviour: The displayed top customers have purchase counts ranging from four to five orders.
8. Revenue Trends: The monthly revenue line chart can be used to identify changes in sales performance over time, once the date axis is sorted correctly.
9. Customer Concentration: The revenue contribution of the top 10 customers can help management assess dependence on a small group of customers.

8. Dashboard Limitations

The current dashboard focuses on sales revenue and customer purchasing behaviour. The supplied Orders dataset does not include actual product cost or profit data, so profit and profit margin cannot be calculated reliably.

Additionally, the monthly revenue chart should be checked for chronological sorting and blank date values before the final submission. Customer rankings should be sorted by purchase count in descending order.

9. Conclusion

The Customer Analysis Dashboard was developed using Microsoft Power BI to analyse customer transactions, product categories, payment methods, monthly revenue, and customer purchasing activity.

The dashboard combines KPI cards and charts to provide a clear overview of business performance. It helps identify popular categories, frequently used payment methods, and customers with high purchase counts. Revenue concentration analysis can further help management understand how revenue is distributed across customers.

Overall, this project demonstrates how Power BI and DAX can transform raw customer and order data into meaningful visual insights that support data-driven decision-making.

10. Learning Outcomes

Through this project, I learned:

1. How to import CSV datasets into Power BI Desktop.

2. How to inspect data types and prepare data for analysis.
3. How to create relationships between customer and order tables.
4. How to create DAX measures for revenue, orders, purchasing customers, and average order value.
5. How to use Card visuals, bar charts, and line charts.
6. How to analyse customer purchase counts and payment preferences.
7. How to build an interactive dashboard for business analysis.
8. How to interpret visual insights to support data-driven decisions.